# Incident Report — Frozen Assets: The Snowflake Tenant Heist

SOC Analysis & Triage · Intermediate Operation

## 1. Header / Metadata

| | |
|---|---|
| **Operation ID** | OPS-2026-0422-10 |
| **Operation name** | Frozen Assets: The Snowflake Tenant Heist |
| **Started (UTC)** | 2026-04-22 01:40:00 |
| **Closed (UTC)** | 2026-04-22 03:16:00 |
| **Analyst** | Ndumiso Mbanjwa |
| **Cohort** | May26 Cybersecurity |
| **Difficulty** | Intermediate |
| **Severity** | Critical |
| **Final verdict** | True Positive |
| **Status** | Closed – Escalated for contractor credential reset, Snowflake network policy and MFA enforcement, and customer-data breach assessment |

## 2. Executive Summary

Lumina Retail Group runs its analytics on a Snowflake data warehouse fronted by single sign-on. Over roughly 96 minutes in the early hours of 22 April 2026, a contractor account, J_HOLLOWAY, was used to sign in from outside the company network without multi-factor authentication, look around the warehouse, and then unload an entire customer table to attacker infrastructure. The scenario's premise is that the password came from an infostealer log — malware output sold on for exactly this purpose — and the audit trail is consistent with that: the first thing the actor did was test the password twice before it worked.

The sequence is short and unambiguous. At 01:40 and 01:41 two logons failed with INCORRECT_USERNAME_PASSWORD from 45.134.142[.]77. At 01:47:12 the same address succeeded with single-factor authentication. The actor then ran three orientation queries — SELECT CURRENT_ROLE(), SHOW DATABASES and SHOW TABLES — which is someone establishing what the stolen identity can reach rather than an analyst doing their job. The session moved to a second address, 146.70.171[.]22, at 02:18, and to a third, 193.32.126[.]40, at 03:02. From that third address the theft happened: a throwaway staging area @LUMINA_PROD.RETAIL.TMP_EXP9920 was created

at 03:03:48, and at 03:05:20 a COPY INTO command unloaded LUMINA_PROD.RETAIL.CUSTOMER_PII into it as gzip-compressed CSV — 4,821,663 customer rows, 2 GiB scanned, run on the ad-hoc warehouse WH_ADHOC. Five GET commands between 03:08 and 03:16 pulled the files down.

Two details make the case. First, every malicious session carried the client application string FROSTHARVEST, which appears nowhere in the tenant's normal activity — the legitimate BI and service accounts use standard drivers and the Snowsight web interface. A single unrecognised client string tied the reconnaissance and the theft to one operator. Second, the three source addresses are all commercial VPN exit nodes (Datacamp, M247 and 31173 Services), and none of them is flagged as malicious by any VirusTotal vendor. Reputation data was of no help here; the case was made entirely on behaviour — no MFA, impossible travel, anomalous client, whole-table unload. This is a confirmed True Positive: all 6 scored tasks were resolved, and the findings rest on the platform's own Snowflake audit records and its ATT&CK Cloud mapping.

## 3. Operation Details

| | |
|---|---|
| **Triggering detection** | Three rules fired: “Snowflake Logon Without MFA From Untrusted IP” (High), “Impossible Travel For Data-Platform Identity” (High) and “Failed Then Successful Logon (Credential Validation)” (Medium) |
| **Source** | External VPN exits 45.134.142[.]77 (foothold), 146.70.171[.]22 (second session) and 193.32.126[.]40 (staging and exfiltration) |
| **Destination** | Lumina Retail Group Snowflake tenant (account xz41928); database LUMINA_PROD, schema RETAIL |
| **User / Host affected** | J_HOLLOWAY (contractor Snowflake user, MFA not used). Baseline accounts LUMINA_BI and SVC_ANALYTICS were unaffected |
| **Artefact** | Client application string FROSTHARVEST; stage @LUMINA_PROD.RETAIL.TMP_EXP9920; table LUMINA_PROD.RETAIL.CUSTOMER_PII; agent snowflake-python-connector/3.10.1; warehouse WH_ADHOC |
| **Direction** | Inbound authentication from external VPN infrastructure; outbound unload of 4,821,663 rows (2 GiB scanned) via stage and GET |
| **Tool surfaces used** | Security log platform (SIEM) and cloud audit telemetry (CLOUD) |

## 4. Investigation & Triage

| Task | Objective | What I checked | Finding | Decision & reasoning | Fig. ref |
|---|---|---|---|---|---|
| Task 1 – Spot the foothold | Identify the account behind the anomaly | Snowflake login and query audit; separated routine BI/service activity from the outlier | J_HOLLOWAY authenticated without MFA from outside corporate egress and then ran orientation queries. LUMINA_BI (10.61.20.40 / 10.61.10.8) and SVC_ANALYTICS account for the normal traffic | Legitimate sessions satisfy MFA and come from corporate addresses; this one did neither, and the follow-on queries show someone orienting rather than working | Fig. 2 |
| Task 2 – Trace the first exit node | Find the earliest evidence of the credential being used from attacker infrastructure | Cloud events filtered to J_HOLLOWAY; cross-checked against the identity-provider sign-in records | 45.134.142[.]77: failed logons at 01:40:00 and 01:41:00 (INCORRECT_USERNAME_PASSWORD), then success at 01:47:12, single-factor, interactive. The AAD sign-in log records the same address | Two failures then a success from one address is password validation — the actor confirming a stolen credential still works. That makes 01:47:12 the start of the intrusion | Fig. 6 |
| Task 3 – Name the tooling tell | Identify the client application unique to the malicious sessions | The clientApplication field recorded on every Snowflake session | FROSTHARVEST appears on the recon queries and on the data theft, and on nothing else in the tenant; normal activity uses standard drivers and Snowsight | Snowflake records which client opened each session. One string that never occurs in baseline activity links every malicious action to a single operator and makes a precise hunting query | Fig. 4 |
| Task 4 – Find the staging step | Locate where the data was gathered | Query history around the exfiltration window | 2026-04-22 03:03:48 “CREATE TEMPORARY STAGE”, category | Bulk theft from a warehouse is normally staged first. A throwaway | Fig. 5 |

| Task | Objective | What I checked | Finding | Decision & reasoning | Fig. ref |
|---|---|---|---|---|---|
| | before download | | Bucket, creating @LUMINA_PROD.RETAIL.TMP_EXP9920; the expanded record shows MFA Not used, agent snowflake-python-connector/3.10.1 and the platform's own T1074 Data Staged tag | stage created minutes before the unload, on an account that has never created one, is the staging step | |
| Task 5 – Recover the exfiltrated table | Trace the COPY INTO back to its source table | The expanded COPY INTO request parameters | 03:05:20 Critical: COPY INTO @LUMINA_PROD.RETAIL.TMP_EXP9920 FROM (SELECT * FROM LUMINA_PROD.RETAIL.CUSTOMER_PII) , FILE_FORMAT= (TYPE=CSV COMPRESSION=GZIP), warehouse WH_ADHOC, bytesScanned 2,147,483,648, rowsUnloaded 4,821,663, clientApplication FROSTHARVEST | The source table is LUMINA_PROD.RETAIL.CUSTOMER_PII, read in full. The row count and the compression settle that this was bulk collection, not a query a person would run | Fig. 3, 4 |
| Task 6 – Classify the exfiltration | Map the exfiltration to the correct ATT&CK technique | The shape of the theft: unload to a stage, then GET to external infrastructure | T1567.002 Exfiltration Over Web Service: Exfiltration to Cloud Storage. Nothing was encrypted in place and no malware C2 channel was involved | The data left through the data platform's own cloud-storage path rather than a malware channel, which is precisely what T1567.002 describes | Fig. 3 |

## 5. Threat Intelligence

| Indicator (defanged) | Tool | Result / score | Date checked (UTC) | Notes |
|---|---|---|---|---|
| 45.134.142[.]77 | VirusTotal | 0 of the security vendors flagged it as malicious (Gridinsoft: Suspicious; all others Clean). AS 212238 Datacamp Limited, network CDNEXT-MIA, 45.134.142.0/24, ARIN, US | 2026-09-19 | Datacamp/CDNEXT is commercial VPN and CDN hosting. A clean verdict is expected for rented egress and is not evidence of safety; this is the address that validated the stolen password and took the foothold. |
| 146.70.171[.]22 | VirusTotal | 0 vendors flagged it (all Unrated/Clean). AS 9009 M247 Europe SRL, 146.70.128.0/18, ARIN, US | 2026-09-19 | M247 is a widely used commercial VPN provider. Its appearance 31 minutes after the first session, on the same identity, is what produced the impossible-travel detection. |
| 193.32.126[.]40 | VirusTotal | 0 vendors flagged it (all Unrated). AS 39351 31173 Services AB, 193.32.126.0/23, RIPE NCC, FR | 2026-09-19 | 31173 Services is Swedish VPN hosting. This is the address the staging and the whole-table unload were run from — the most damaging activity came from the least remarkable-looking address. |
| FROSTHARVEST (client application string) | Internal telemetry | Not a public indicator — no reputation source applies | 2026-09-19 | Its value is local: it appears on every malicious session and on no legitimate one, so it is the highest-quality hunting indicator this case produced. |
| 10.61.20.40 and 10.61.10.8 | VirusTotal | Not applicable — private (internal) | 2026-09-19 | Corporate egress used by LUMINA_BI and |

| Indicator (defanged) | Tool | Result / score | Date checked (UTC) | Notes |
|---|---|---|---|---|
| | | addresses | | SVC_ANALYTICS. Listed only to record the baseline the intrusion was judged against. |

This is assessed with high confidence as credential-driven data theft from a cloud data platform, consistent with the infostealer-log economy: a contractor's Snowflake password is stolen by commodity malware, sold on, validated, and used from rented VPN exits against a tenant with neither MFA nor a network policy. External intelligence contributed nothing to the verdict — all three addresses are unflagged by every VirusTotal vendor, which is the normal state of commercial VPN infrastructure and is treated here as inconclusive rather than exonerating. The evidence that carried the case is entirely internal and behavioural: a failed-then-successful logon pattern, single-factor authentication from outside corporate egress, three source addresses on one identity inside 76 minutes, a client application string that appears nowhere in the baseline, and a whole-table unload of 4.8 million rows to a stage created 92 seconds earlier.

## 6. Timeline (UTC)

| Time (UTC) | Event |
|---|---|
| 2026-04-22 01:40:00 | Failed Snowflake logon for J_HOLLOWAY from 45.134.142[.]77 — INCORRECT_USERNAME_PASSWORD |
| 2026-04-22 01:41:00 | Second failed logon from the same address — the actor testing the stolen credential |
| 2026-04-22 01:47:12 | High — successful ConsoleLogin for J_HOLLOWAY from 45.134.142[.]77, single-factor, no MFA (T1078) |
| 2026-04-22 01:49:00 | Query: SELECT CURRENT_ROLE() — establishing what the stolen identity can do (T1087) |
| 2026-04-22 01:50:00 | Query: SHOW DATABASES — schema enumeration (T1580) |
| 2026-04-22 01:51:00 | Query: SHOW TABLES — mapping what is reachable (T1580) |
| 2026-04-22 02:18:40 | High — second successful logon from 146.70.171[.]22, no MFA, new client — triggers impossible travel |
| 2026-04-22 02:21:05 | High — Query: SELECT * against the target data (T1213) |
| 2026-04-22 03:02:11 | High — third successful logon, from 193.32.126[.]40, no MFA |
| 2026-04-22 03:03:48 | High — CREATE TEMPORARY STAGE @LUMINA_PROD.RETAIL.TMP_EXP9920 (T1074 Data Staged) |
| 2026-04-22 03:05:20 | Critical — COPY INTO unloads LUMINA_PROD.RETAIL.CUSTOMER_PII to the stage: 4,821,663 rows, |

| Time (UTC) | Event |
|---|---|
| | 2 GiB scanned, gzip CSV, client FROSTHARVEST (T1530) |
| 2026-04-22 03:08:00–03:16:00 | Critical — five GET commands retrieve the staged files to external infrastructure (T1567.002) |

## 7. Indicators of Compromise

| Type | Indicator (defanged) | Context |
|---|---|---|
| Account | J_HOLLOWAY | Contractor Snowflake user (account xz41928) whose credential was replayed — platform Confirmed finding |
| IPv4 | 45.134.142[.]77 | First exit node: two failed logons then the no-MFA success at 01:47:12 — platform Confirmed finding |
| IPv4 | 146.70.171[.]22 | Second exit node (02:18:40); source of the impossible-travel detection |
| IPv4 | 193.32.126[.]40 | Third exit node; staging, COPY INTO and all five GET retrievals |
| Client application | FROSTHARVEST | Client string on every malicious session, absent from all baseline activity — platform Confirmed finding |
| Snowflake stage | @LUMINA_PROD.RETAIL.TMP_EXP9920 | Attacker-created temporary stage holding the compressed dump — platform Confirmed finding |
| Database table | LUMINA_PROD.RETAIL.CUSTOMER_PII | Table unloaded in full, 4,821,663 rows — platform Confirmed finding |
| Warehouse | WH_ADHOC | Compute warehouse used for the unload |
| User agent | snowflake-python-connector/3.10.1 | Agent recorded on the staging and exfiltration requests |
| Error code | INCORRECT_USERNAME_PASSWORD | Recorded twice at 01:40 and 01:41 — credential validation before success |

### STIX 2.1 Bundle


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SOC Simulator operation Frozen Assets: The Snowflake Tenant Heist.",
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tenant.",
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```

```
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stage.",
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Limited (2026-09-19).",
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                "marking-definition--94868c89-83c2-464b-929b-a1a8aa3c8487"
            ]
        },
        {
            "type": "indicator",
            "spec_version": "2.1",
            "id": "indicator--9e66cab2-b583-5c14-85ed-d9df9232a9f2",
            "created": "2026-04-22T01:40:00.000Z",
            "modified": "2026-09-19T14:30:00.000Z",
```

```
            "name": "Anomalous Snowflake client application",
            "description": "Client application string recorded on all malicious sessions;
never seen in baseline tenant activity.",
            "indicator_types": [
                "malicious-activity"
            ],
            "pattern": "[software:name = 'FROSTHARVEST']",
            "pattern_type": "stix",
            "valid_from": "2026-04-22T01:40:00.000Z",
            "created_by_ref": "identity--f875f44b-7984-5af6-b39a-255c1c42c3e6",
            "object_marking_refs": [
                "marking-definition--94868c89-83c2-464b-929b-a1a8aa3c8487"
            ]
        },
        {
            "type": "indicator",
            "spec_version": "2.1",
            "id": "indicator--2eb3e6a1-60c6-539d-8631-d820bf04f9f2",
            "created": "2026-04-22T01:40:00.000Z",
            "modified": "2026-09-19T14:30:00.000Z",
            "name": "Attacker-created staging area",
            "description": "Temporary Snowflake stage created 2026-04-22T03:03:48Z to hold
the gzip CSV dump.",
            "indicator_types": [
                "malicious-activity"
            ],
            "pattern": "[directory:path = '@LUMINA_PROD.RETAIL.TMP_EXP9920']",
            "pattern_type": "stix",
            "valid_from": "2026-04-22T01:40:00.000Z",
            "created_by_ref": "identity--f875f44b-7984-5af6-b39a-255c1c42c3e6",
            "object_marking_refs": [
                "marking-definition--94868c89-83c2-464b-929b-a1a8aa3c8487"
            ]
        },
        {
            "type": "indicator",
            "spec_version": "2.1",
            "id": "indicator--d6ea9741-61dc-5bee-b539-5a511691268f",
            "created": "2026-04-22T01:40:00.000Z",
            "modified": "2026-09-19T14:30:00.000Z",
            "name": "Exfiltrated customer table",
            "description": "LUMINA_PROD.RETAIL.CUSTOMER_PII unloaded in full at 2026-04-
22T03:05:20Z: 4,821,663 rows, 2,147,483,648 bytes scanned.",
            "indicator_types": [
                "malicious-activity"
            ],
            "pattern": "[file:name = 'CUSTOMER_PII']",
            "pattern_type": "stix",
            "valid_from": "2026-04-22T01:40:00.000Z",
            "created_by_ref": "identity--f875f44b-7984-5af6-b39a-255c1c42c3e6",
            "object_marking_refs": [
                "marking-definition--94868c89-83c2-464b-929b-a1a8aa3c8487"
            ]
        }
    ]
}
```

## 8. MITRE ATT&CK Mapping

| Tactic | Technique | ID |
|---|---|---|
| Initial Access, Persistence, Privilege Escalation, Stealth | Valid Accounts | T1078 |
| Discovery | Account Discovery | T1087 |
| Discovery | Cloud Infrastructure Discovery | T1580 |
| Collection | Data from Information Repositories | T1213 |
| Collection | Data Staged | T1074 |
| Collection | Data from Cloud Storage | T1530 |
| Exfiltration | Exfiltration Over Web Service: Exfiltration to Cloud Storage | T1567.002 |

T1078, T1087, T1580, T1213, T1074 and T1530 are taken directly from the platform's “MITRE ATT&CK Cloud — techniques used in this incident” panel, with the tactic placement shown there; T1074 is additionally tagged on the CREATE TEMPORARY STAGE event itself (Fig. 5). T1567.002 is the platform's own confirmed answer to the scored classification task. No technique has been added beyond what the platform confirms.

## 9. Impact & Containment

Impact: an entire customer table — 4,821,663 rows of personally identifiable information, 2 GiB scanned and compressed into gzip CSV — left the production Snowflake tenant and was retrieved to attacker-controlled infrastructure. The tenant had no MFA requirement and no network policy on the contractor identity, so a single stolen password was sufficient; nothing between the credential and the data stopped the actor. Anything reachable by J_HOLLOWAY's role must be treated as exposed, not just the table that was unloaded. Containment steps, in priority order:

1. Disable J_HOLLOWAY immediately and revoke all active Snowflake sessions and tokens for the account; reset the credential and treat the old password as public.
2. Enforce MFA on every human Snowflake identity, and apply a network policy restricting logon to corporate egress ranges — either control alone would have stopped this intrusion.
3. Drop the stage @LUMINA_PROD.RETAIL.TMP_EXP9920 after preserving evidence, and audit all stages and GET activity for other unloads not covered by this incident.
4. Block 45.134.142[.]77, 146.70.171[.]22 and 193.32.126[.]40, and hunt the full query history for the client application string FROSTHARVEST to confirm the intrusion is bounded by these 17 events.
5. Engage the data owners and privacy/legal function on breach assessment for 4,821,663 customer records, and review which roles can read CUSTOMER_PII in full at all.

## 10. Conclusion & Recommendations

Final verdict: True Positive. A contractor credential, consistent with the scenario's infostealer-log origin, was validated with two failed logons and then used without MFA from three commercial VPN

exits over 96 minutes. The actor oriented itself with three standard Snowflake commands, created a throwaway stage, unloaded 4,821,663 customer records as compressed CSV and pulled them down in five retrievals. All 6 scored tasks were resolved and every finding is backed by the platform's audit records and its own ATT&CK Cloud mapping.

The lesson worth carrying out of this case is where the detection value actually sat. External reputation data was silent: not one VirusTotal vendor flags any of the three addresses, because renting a VPN exit is not itself malicious. What exposed the intrusion was the behaviour around the identity — authentication that skipped MFA, the same account appearing from distant addresses within the hour, a client application string that existed nowhere else in the tenant, and a single query that read a customer table in full. A SOC that only alerts on known-bad infrastructure would have missed this entirely.

Recommendations:

1. Require MFA and a network policy for every non-service Snowflake identity, and review contractor accounts specifically — they are frequently exempted from controls applied to staff.
2. Alert on any client application string that has not been seen in the tenant before; in this case that one signal covers the whole intrusion from recon to exfiltration.
3. Alert on stage creation followed closely by COPY INTO, and on any single query whose rowsUnloaded or bytesScanned exceeds a normal-day threshold for the warehouse.
4. Treat failed-then-successful authentication on one identity as a credential-validation signal and raise it for review, rather than clearing it because the eventual logon succeeded.
5. Monitor infostealer-log marketplaces for corporate and contractor identities, and force resets when company credentials appear in stealer dumps.
6. Restrict ad-hoc warehouses such as WH_ADHOC and full-table reads of PII tables to named roles, with logging reviewed on a schedule rather than only after an alert.

## 11. Evidence / Screenshots

| Figure | Caption | Referenced in |
|---|---|---|
| Figure 1 | The completed operation console showing 6/6 scored tasks and the three detection rules the platform fired for this incident | Sections 3, 4 |
| Figure 2 | SIEM security logs (102 entries) alongside the confirmed finding for task 1, the compromised contractor account J_HOLLOWAY | Task 1 |
| Figure 3 | Snowflake cloud events filtered to J_HOLLOWAY — 17 of 185 events — showing the COPY INTO at 03:05:20 and the five critical GET retrievals that follow | Tasks 5, 6 |
| Figure 4 | The expanded COPY INTO request parameters: source table CUSTOMER_PII, gzip CSV, warehouse WH_ADHOC, 2 GiB scanned, 4,821,663 rows unloaded, client application FROSTHARVEST | Tasks 3, 5 |
| Figure 5 | The CREATE TEMPORARY STAGE event expanded, showing MFA not used, the snowflake-python-connector agent, the stage @LUMINA_PROD.RETAIL.TMP_EXP9920 and the platform's T1074 Data Staged tag | Task 4 |
| Figure 6 | The foothold sequence from 45.134.142[.]77: two failed logons at 01:40 and 01:41, the single-factor success at 01:47:12, then the three reconnaissance queries | Tasks 1, 2 |
| Figure 7 | The platform's resource-impact panel, recording lumina-retail as compromised by a no-MFA contractor logon with an unrecognised client application, and CUSTOMER_PII as critical | Sections 8, 9 |

**Figure 1. The completed operation console showing 6/6 scored tasks and the three detection rules the platform fired for this incident.**

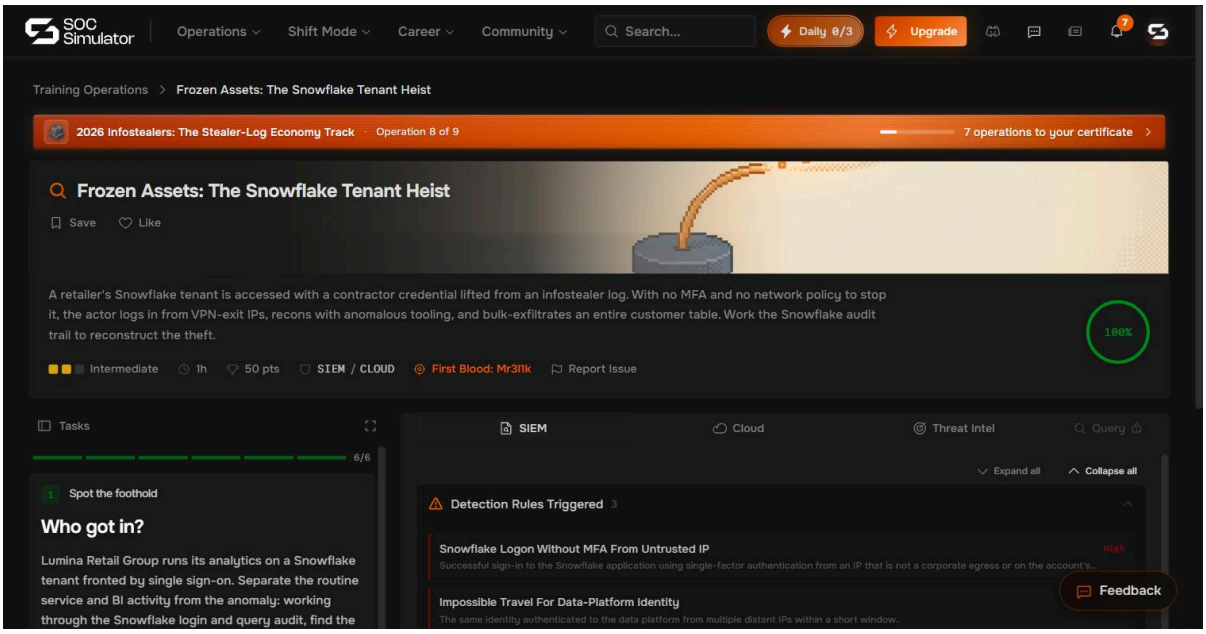



**Figure 2. SIEM security logs (102 entries) alongside the confirmed finding for task 1, the compromised contractor account J_HOLLOWAY.**

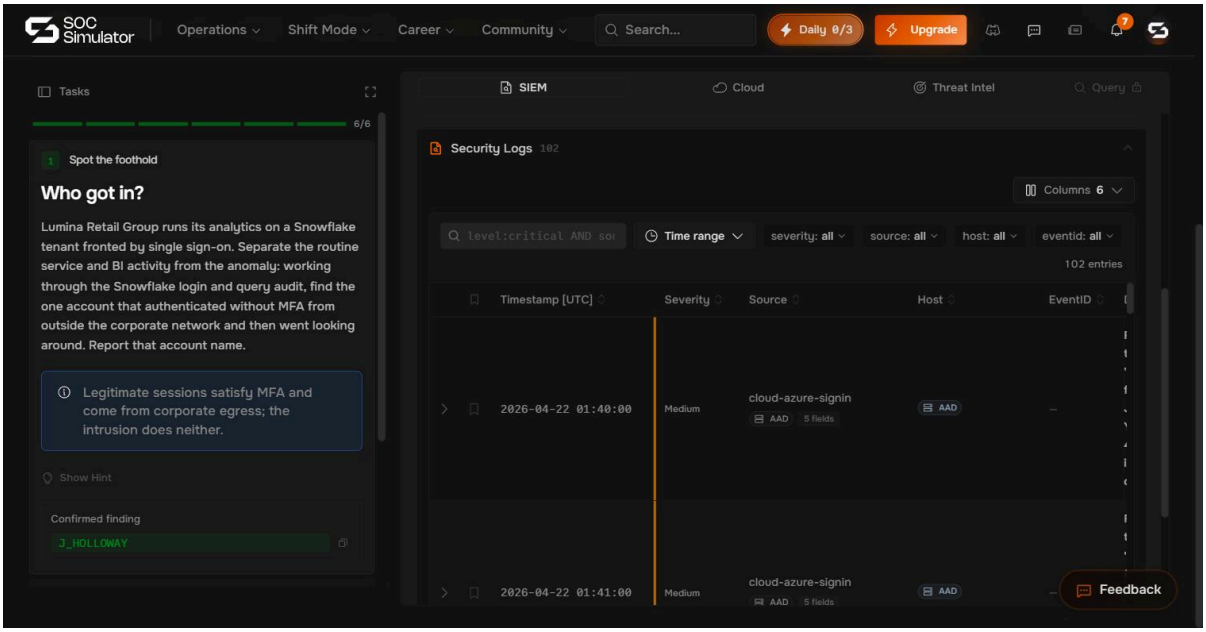


**Figure 3. Snowflake cloud events filtered to J_HOLLOWAY — 17 of 185 events — showing the COPY INTO at 03:05:20 and the five critical GET retrievals that follow.**

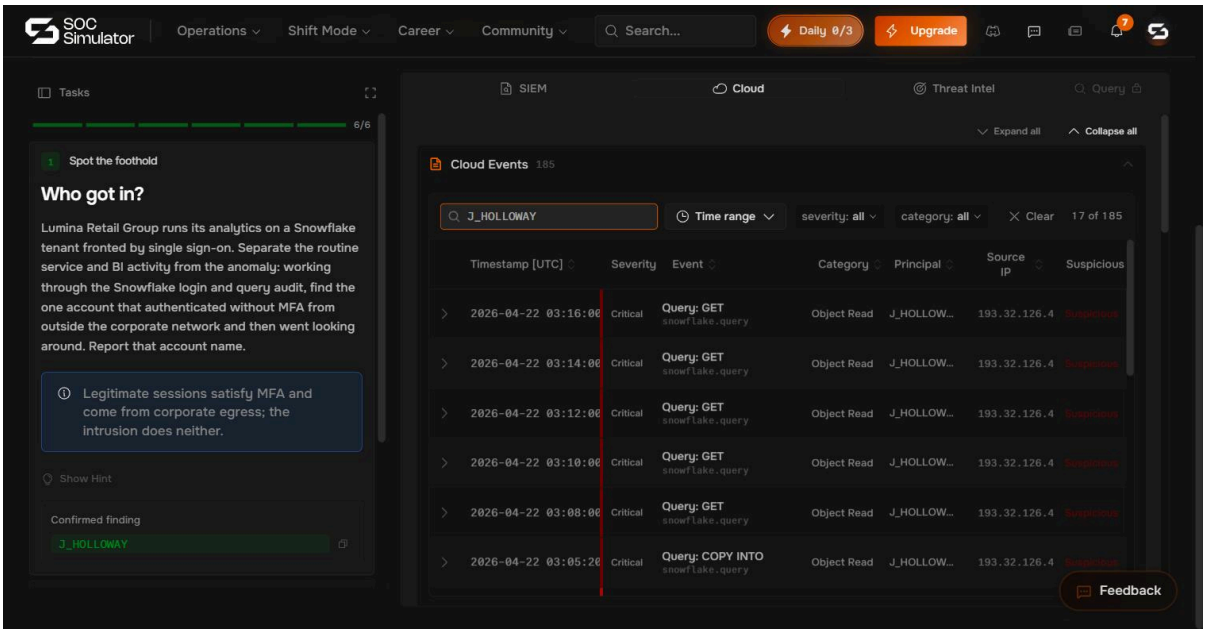



**Figure 4. The expanded COPY INTO request parameters: source table CUSTOMER_PII, gzip CSV, warehouse WH_ADHOC, 2 GiB scanned, 4,821,663 rows unloaded, client application FROSTHARVEST.**

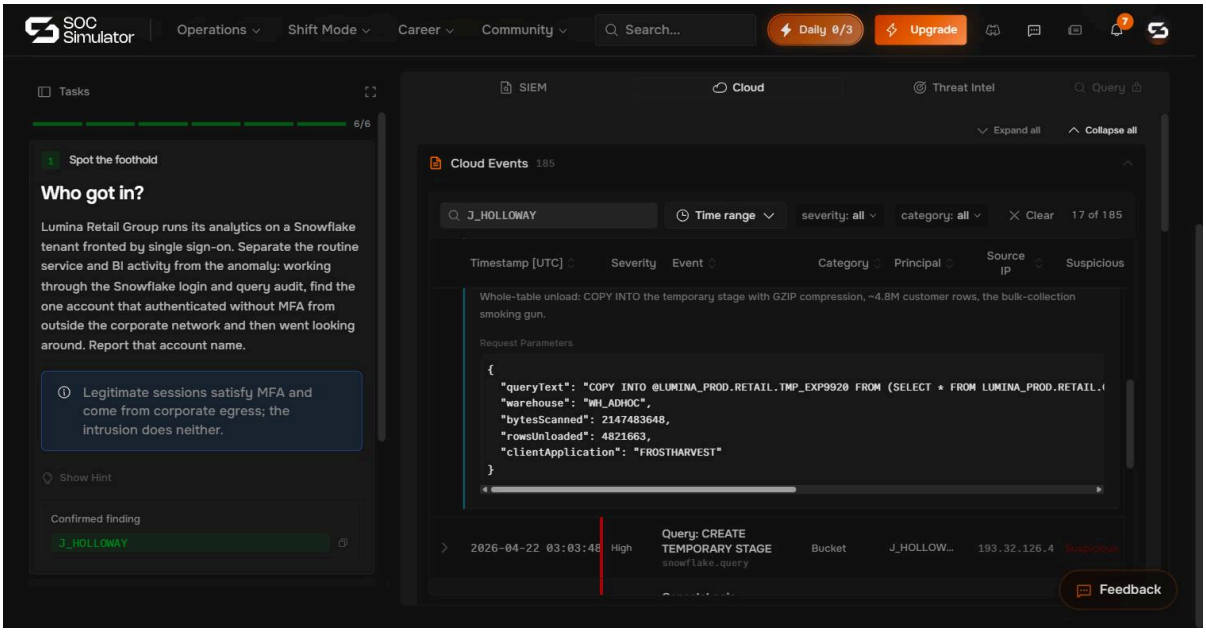


**Figure 5. The CREATE TEMPORARY STAGE event expanded, showing MFA not used, the snowflake-python-connector agent, the stage @LUMINA_PROD.RETAIL.TMP_EXP9920 and the platform's T1074 Data Staged tag.**

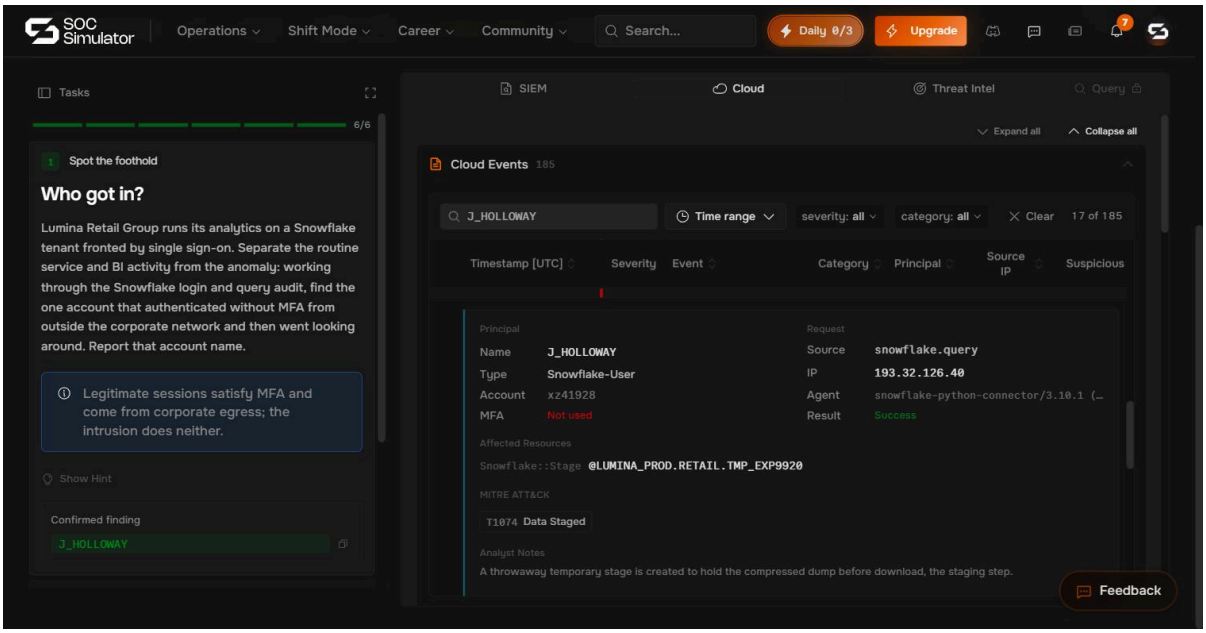



**Figure 6. The foothold sequence from 45.134.142[.]77: two failed logons at 01:40 and 01:41, the single-factor success at 01:47:12, then the three reconnaissance queries.**

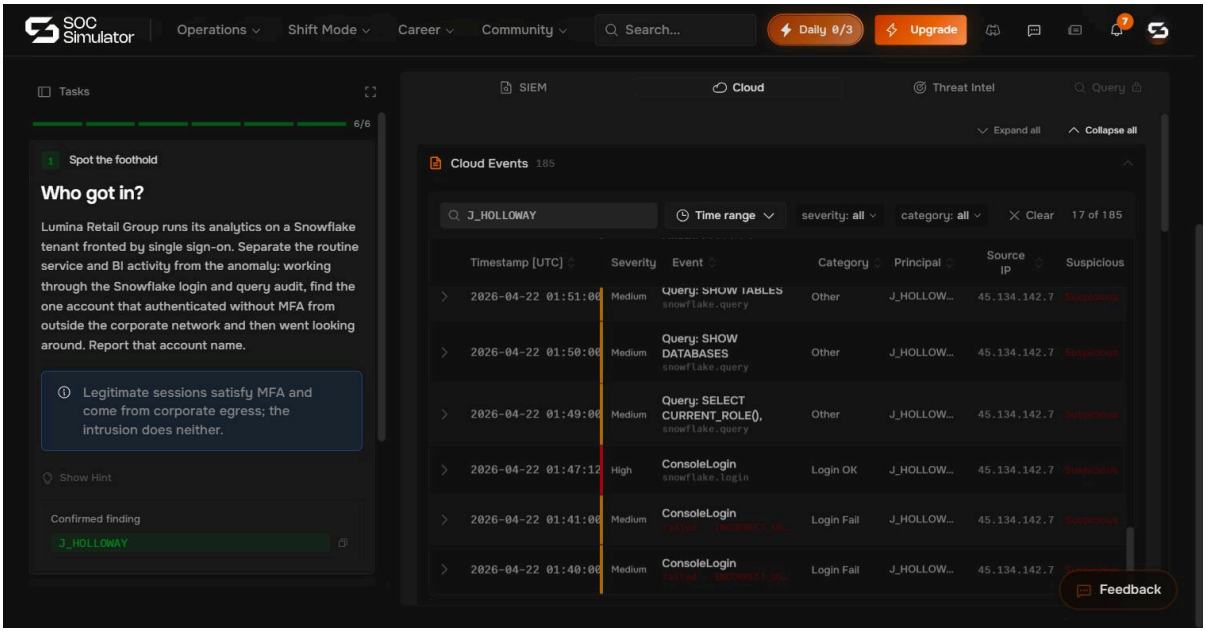


**Figure 7. The platform's resource-impact panel, recording lumina-retail as compromised by a no-MFA contractor logon with an unrecognised client application, and CUSTOMER_PII as critical.**

